DEPARTAMENTO DE
MATEMÁTICA

The Principles of Deepn Learning Theory
Capítulo 4: RG-flow de pré-ativações

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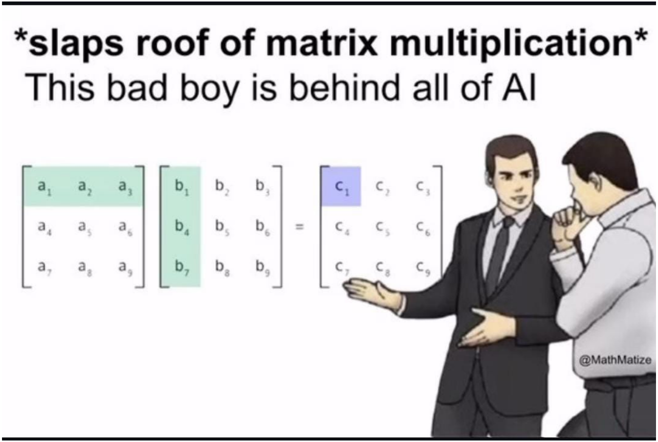
slaps roof of matrix multiplication
This bad boy is behind all of AI
a1 a2 a3
a4 a5 a6
a7 a8 a9
b1 b2 b3
b4 b5 b6
b7 b8 b9
=
c1 c2 c3
c4 c5 c6
c7 c8 c9
@MathMatize

Dataset com inputs n_0-dimensionais e com N_D amostras:

$$D = \{x_{i,\alpha}\}_{i=1,\ldots,n_0;\alpha=1,\ldots,N_D}$$

Pré-ativações da 1ª camada são dadas por:

$$z_{i,\alpha}^{(1)} \equiv z_i^{(1)}(x_\alpha) = b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha}, \quad \text{para } i = 1, \ldots, n_1.$$

Na inicialização, estabelecemos $b^{(1)}, W^{(1)}$ independentemente distribuídas e tais que:

$$\mathbb{E}[b_i^{(1)}] = \mathbb{E}[W_{i_1 j_1}^{(1)}] = 0$$

e

$$\mathbb{E}[b_i^{(1)} b_j^{(1)}] = \delta_i j C_b^{(1)}$$

$$\mathbb{E}[W_{i_1 j_1}^{(1)} W_{i_2 j_2}^{(1)}] = \delta_{i_1 i_2} \delta_{j_1 j_2} \frac{C_W^{(1)}}{n_0}$$

Estamos interessados em:

$$p(z^{(1)}|D) = p(z^{(1)}(x_1), \ldots, z^{(1)}(x_{N_D}))$$

Vamos fazer a demonstração via 2 caminhos:

- combinatório, via contrações de Wick e correlatores;
- algébrico, via transformada de Hubbard-Stratonovich.

Vemos primeiramente que:

$$\mathbb{E}[z_{i,\alpha}^{(1)}] = \mathbb{E}[b_i^{(1)} + \sum_{j=1}^{n_0} W_{ij}^{(1)} x_{j,\alpha_1}]$$
$$= \mathbb{E}[b_i^{(1)}] + \sum_{j=1}^{n_0} \mathbb{E}[W_{ij}] x_{j,\alpha} = 0$$

dado que as distribuições tem média 0.
É ”fácil ver que”todos os correlatores de ordem ímpar de $p(z^{(1)}|D)$ são zerados, porque sempre sobra um $b^{(1)}$ ou $W^{(1)}$ em cada termo que zera a conta.

Para um correlator de dois pontos:

$$\mathbb{E}\left[z^{(1)}_{i_1;\alpha_1}\, z^{(1)}_{i_2;\alpha_2}\right] = \mathbb{E}\left[\left(b^{(1)}_{i_1} + \sum_{j_1=1}^{n_0} W^{(1)}_{i_1 j_1} x_{j_1;\alpha_1}\right)\left(b^{(1)}_{i_2} + \sum_{j_2=1}^{n_0} W^{(1)}_{i_2 j_2} x_{j_2;\alpha_2}\right)\right]$$

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$$= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1,j_1}] x_{j_1,\alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2,j_2}] x_{j_2,\alpha_2}$$

$$+ \mathbb{E}[(\sum_{j_1=1}^{n_0} W^{(1)}_{i_1 j_1} x_{j_1;\alpha_1})(\sum_{j_2=1}^{n_0} W^{(1)}_{i_2 j_2} x_{j_2;\alpha_2})]$$

Para um correlator de dois pontos:

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\begin{aligned}
\mathbb{E}\left[z^{(1)}_{i_1;\alpha_1}\, z^{(1)}_{i_2;\alpha_2}\right] &= \mathbb{E}\left[\left(b^{(1)}_{i_1} + \sum_{j_1=1}^{n_0} W^{(1)}_{i_1 j_1} x_{j_1;\alpha_1}\right)\left(b^{(1)}_{i_2} + \sum_{j_2=1}^{n_0} W^{(1)}_{i_2 j_2} x_{j_2;\alpha_2}\right)\right] \\
&= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1,j_1}] x_{j_1,\alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2,j_2}] x_{j_2,\alpha_2} \\
&\quad + \mathbb{E}[(\sum_{j_1=1}^{n_0} W^{(1)}_{i_1 j_1} x_{j_1;\alpha_1})(\sum_{j_2=1}^{n_0} W^{(1)}_{i_2 j_2} x_{j_2;\alpha_2})] \\
&= \delta_{i_1 i_2} C^{(1)}_b + \sum_{j_1,j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j,\alpha_1} x_{j,\alpha_2}
\end{aligned}
$$

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&= \mathbb{E}[b_{i_1} b_{i_2}] + \sum_{j_1} \mathbb{E}[b_{i_2} W_{i_1,j_1}] x_{j_1,\alpha_1} + \sum_{j_2} \mathbb{E}[b_{i_1} W_{i_2,j_2}] x_{j_2,\alpha_2} \\
&\quad + \mathbb{E}[(\sum_{j_1=1}^{n_0} W^{(1)}_{i_1 j_1} x_{j_1;\alpha_1})(\sum_{j_2=1}^{n_0} W^{(1)}_{i_2 j_2} x_{j_2;\alpha_2})] \\
&= \delta_{i_1 i_2} C^{(1)}_b + \sum_{j_1,j_2=1}^{n_0} \mathbb{E}[W_{i_1 j_1} W_{i_2 j_2}] x_{j,\alpha_1} x_{j,\alpha_2} \\
&= \delta_{i_1 i_2}\left(C^{(1)}_b + C^{(1)}_W \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j;\alpha_1} x_{j;\alpha_2}\right)
\end{aligned}
$$

Definimos a seguinte expressão como a **métrica de primeira camada**:

$$G^{(1)}_{\alpha_1\alpha_2} := C^{(1)}_b + C^{(1)}_W \frac{1}{n_0} \sum_{j=1}^{n_0} x_{j;\alpha_1} x_{j;\alpha_2}$$

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Note que é uma função de $x_{\alpha_1}, x_{\alpha_2}$. Assim, nossa igualdade acima se traduz como:

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$$\mathbb{E}\left[z^{(1)}_{i_1;\alpha_1}\, z^{(1)}_{i_2;\alpha_2} \right] = \delta_{i_1 i_2} G^{(1)}_{\alpha_1\alpha_2}$$

Seguindo a mesma ideia:

$$\mathbb{E}[z^{(1)}_{i_1;\alpha_1} z^{(1)}_{i_2;\alpha_2} z^{(1)}_{i_3;\alpha_3} z^{(1)}_{i_4;\alpha_4}] = \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(1)}_{\alpha_1\alpha_2} G^{(1)}_{\alpha_3\alpha_4} + \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(1)}_{\alpha_1\alpha_3} G^{(1)}_{\alpha_2\alpha_4} + \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(1)}_{\alpha_1\alpha_4} G^{(1)}_{\alpha_2\alpha_3}$$

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Lembrando de discussão anterior, isso significa que

$$\mathbb{E}[z^{(1)}_{i_1;\alpha_1} z^{(1)}_{i_2;\alpha_2} z^{(1)}_{i_3;\alpha_3} z^{(1)}_{i_4;\alpha_4}]\Big|_{\text{connected}} = 0$$

de modo que a distribuição é Gaussiana.

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De forma similar, podemos verificar que todos os correlatores conexos de ordem mais alta zeram, e portanto, podem ser gerados por uma Gaussiana.

Lembrando, uma distribuição Gaussiana multivariada z de média s e matriz de covariância K pode ser escrita como

$$
\begin{aligned}
p(z) &= \frac{1}{\sqrt{|2\pi K|}} \exp\left[-\frac{1}{2}\sum_{\mu,\nu=1}^{N}(z-s)_\mu K^{\mu\nu}(z-s)_\nu\right] \\
&= \frac{1}{\sqrt{|2\pi K|}} \exp\left[-\frac{1}{2}(z-s)^T K^{-1}(z-s)\right]
\end{aligned}
$$

onde $K^{\mu\nu} = (K^{-1})_{\mu\nu}$.

Para escrever a distribuição de primeira camada, precisamos da inversa de $\delta_{i_1 i_2} G^{\alpha_1 \alpha_2}_{(1)}$, para cada par de índices i_1, i_2, que satisfaça

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$$\sum_{j=1}^{n_1} \sum_{\beta \in D} (\delta_{i_1 j} G_{(1)}^{\alpha_1 \beta})(\delta_{i_2 j} G^{(1)}_{\beta \alpha_2}) = \delta_{i_1 i_2} \delta^{\alpha_1}_{\alpha_2}$$

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$G_{(1)}^{\alpha\beta}$ denota as entradas da matriz $(G^{(1)})^{-1}_{\alpha\beta} \in \mathbb{R}^{N_D}$, com as métricas $G^{(1)}_{\alpha\beta}$ para duas entradas x_α, x_β, satisfazendo

$$\sum_{\beta \in D} G_{(1)}^{\alpha_1 \beta} G^{(1)}_{\beta \alpha_2} = \delta^{\alpha_1}_{\alpha_2}$$

Podemos, então, escrever a distribuição $p(z^{(1)}|D)$ na forma:

$$p(z^{(1)}|D) = \frac{1}{Z}e^{-S(z^{(1)})}$$

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$$S(z^{(1)}) = \frac{1}{2} \sum_{i=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z_{i;\alpha_1}^{(1)} G_{(1)}^{\alpha_1 \alpha_2} z_{i;\alpha_2}^{(1)}$$

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$$\left(= \sum_{i,j=1}^{n_1} \sum_{\alpha_1, \alpha_2 \in D} z^{(1)}_{i;\alpha_1} \delta_{ij} G^{\alpha_1 \alpha_2}_{(1)} z^{(1)}_{j;\alpha_2} \right)$$

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e Z é a *função de partição*:

$$Z = \int \left[\prod_{i,\alpha} dz^{(1)}_{i,\alpha} \right] e^{-S(z^{(1)})} = |2\pi G^{(1)}|^{\frac{n_1}{2}}$$

Como visto no capítulo anterior, podemos expressar a distribuição na forma

$$p(z \mid D) = \int \left[\prod_j db_j \, p(b_j)\right]\left[\prod_{j,k} dW_{jk} \, p(W_{jk})\right]\prod_{j,\alpha} \delta\left(z_{j;\alpha} - b_i - \sum_j W_{jk} x_{j;\alpha}\right)$$

suprimindo por hora os sobrescritos (1).

Por hipótese:

$$p(b_j) = \frac{1}{\sqrt{2\pi C_b}} \exp(-\frac{b_j^2}{2C_b})$$

$$p(W_{jk}) = \frac{1}{\sqrt{2\pi C_W/n_0}} \exp(-\frac{n_0 W_{jk}^2}{2C_W})$$

Vamos usar a **transformação de Hubbard-Stratonovich**.
I.e., a representação integral do delta de Dirac:

$$\delta(z-a) = \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)}$$

Obtemos:

$$p(z \mid D) = \int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}} \exp(-\frac{b_j^2}{2C_b})\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}} \exp(-\frac{n_0 W_{jk}^2}{2C_W})\right]\left[\prod_{j,\alpha} \int \frac{d\Lambda}{2\pi} e^{i\Lambda(z-a)}\right]$$

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$$= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}}\right]\left[\prod_{i,j} \frac{dW_{ij}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{i,\alpha} \frac{d\Lambda_i^\alpha}{2\pi}\right]$$

$$\times \exp\left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha}\right)\right]$$

Completando quadrados:

$$\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{i,j} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{k;\alpha} \right)$$

$$= -\frac{1}{2C_b} \sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha \right)^2 - \frac{n_0}{2C_W} \sum_{j,k} \left(W_{jk} + i \frac{C_W}{n_0} \sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2$$

$$- \frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha}$$

Voltando à exponencial,

$$\int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}}\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right]$$

$$\times \exp\left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i\sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha}\right)\right]$$

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\begin{aligned}
&\int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}}\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right] \\
&\times \exp\left[-\sum_j \frac{b_j^2}{2C_b} - n_0 \sum_{jk} \frac{W_{jk}^2}{2C_W} + i \sum_{j,\alpha} \Lambda_j^\alpha \left(z_{j;\alpha} - b_j - \sum_j W_{jk} x_{j;\alpha}\right)\right] \\
&= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}}\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right] \\
&\exp\left[-\frac{1}{2C_b}\sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha\right)^2\right] \exp\left[-\frac{n_0}{2C_W}\sum_{j,k}\left(W_{jk} + i\frac{C_W}{n_0}\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2\right] \\
&\exp\left[-\frac{C_W}{2n_0}\sum_{j,k}\left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2 - \frac{C_b}{2}\sum_j\left(\sum_\alpha \Lambda_j^\alpha\right)^2 + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}\right]
\end{aligned}
$$

$$= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}}\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right]$$

$$\exp\left[-\frac{1}{2C_b}\sum_j\left(b_j + iC_b\sum_\alpha \Lambda_j^\alpha\right)^2\right]\exp\left[-\frac{n_0}{2C_W}\sum_{j,k}\left(W_{jk} + i\frac{C_W}{n_0}\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2\right]$$

$$\exp\left[-\frac{C_W}{2n_0}\sum_{j,k}\left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2 - \frac{C_b}{2}\sum_j\left(\sum_\alpha \Lambda_j^\alpha\right)^2 + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}\right]$$

$$= \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}}\right]\left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right]$$

$$\exp\left[-\frac{1}{2C_b}\sum_j\left(b_j + iC_b\sum_\alpha \Lambda_j^\alpha\right)^2\right]\exp\left[-\frac{n_0}{2C_W}\sum_{j,k}\left(W_{jk} + i\frac{C_W}{n_0}\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2\right]$$

$$\exp\left[-\frac{C_W}{2n_0}\sum_{j,k}\left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2 - \frac{C_b}{2}\sum_j\left(\sum_\alpha \Lambda_j^\alpha\right)^2 + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}\right]$$

$$\times = \int \left[\prod_j \frac{db_i}{\sqrt{2\pi C_b}}\right]\exp\left[-\frac{1}{2C_b}\sum_j\left(b_j + iC_b\sum_\alpha \Lambda_j^\alpha\right)^2\right]$$

$$\times \int \left[\prod_{j,k} \frac{dW_{jk}}{\sqrt{2\pi C_W/n_0}}\right]\exp\left[-\frac{n_0}{2C_W}\sum_{j,k}\left(W_{jk} + i\frac{C_W}{n_0}\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2\right]$$

$$\int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi}\right]\exp\left[-\frac{C_W}{2n_0}\sum_{j,k}\left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha}\right)^2 - \frac{C_b}{2}\sum_j\left(\sum_\alpha \Lambda_j^\alpha\right)^2 + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}\right]$$

Olhando bem, são integrais Gaussianas deslocadas por uma média:

$$\int \left[\prod_j \frac{db_j}{\sqrt{2\pi C_b}}\right] \exp\left[-\frac{1}{2C_b}\sum_j \left(b_j + iC_b \sum_\alpha \Lambda_j^\alpha\right)^2\right]$$
$$= \prod_j \int \left[\frac{db_j}{\sqrt{2\pi C_b}} \exp\left(-\frac{(b_j - \mu_b)^2}{2C_b}\right)\right]$$

com a média $\mu_b = -iC_b \sum_\alpha \Lambda_j^\alpha$. Assim, cada integral dessa é igual a 1, e some da representação.

Olhando bem, são integrais Gaussianas deslocadas por uma média:

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com a média $\mu_b = -iC_b \sum_\alpha \Lambda_j^\alpha$. Assim, cada integral dessa é igual a 1, e some da representação.

De forma análoga, a integral envolvendo W_{jk} vira 1.

Então...

$$p(z \mid D) = \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp\left[-\frac{C_W}{2n_0} \sum_{j,k} \left(\sum_\alpha \Lambda_j^\alpha x_{k;\alpha} \right)^2 - \frac{C_b}{2} \sum_j \left(\sum_\alpha \Lambda_j^\alpha \right)^2 + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]$$

$$= \int \left[\prod_{j,\alpha} \frac{d\Lambda_j^\alpha}{2\pi} \right] \exp\left[-\frac{C_W}{2n_0} \sum_{j,k} \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} x_{k;\alpha_1} x_{k;\alpha_2} - \frac{C_b}{2} \sum_j \sum_{\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]$$

$$= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp\left[-\frac{1}{2} \sum_{j,\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} \left(C_b + C_W \frac{1}{n_0} \sum_k x_{k;\alpha_1} x_{k;\alpha_2} \right) + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]$$

$$= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi} \right] \exp\left[-\frac{1}{2} \sum_{j,\alpha_1 \alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1 \alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \right]$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha); z = (z_{j;\alpha}); G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$-\frac{1}{2} \sum_{j,\alpha_1\alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1\alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha}$$

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$$= -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right]$$

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$$-\frac{1}{2}\sum_{j,\alpha_1\alpha_2} \Lambda_j^{\alpha_1}\Lambda_j^{\alpha_2} G_{\alpha_1\alpha_2} + i\sum_{j,\alpha} \Lambda_j^{\alpha} z_{j;\alpha}$$

$$= -\frac{1}{2}\left[\Lambda^T G \Lambda - 2i\Lambda^T z\right]$$

$$= -\frac{1}{2}\left[\Lambda^T G \Lambda - i\Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda\right]$$

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$$-\frac{1}{2}\sum_{j,\alpha_1\alpha_2} \Lambda_j^{\alpha_1}\Lambda_j^{\alpha_2} G_{\alpha_1\alpha_2} + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}$$

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$$= -\frac{1}{2}\left[\Lambda^T G\Lambda - i\Lambda^T G G^{-1} z - i z^T G^{-T} G\Lambda\right]$$

$$= -\frac{1}{2}\left[\Lambda^T G\Lambda - i\Lambda^T G G^{-1} z - i(G^{-1}z)^T G\Lambda - (G^{-1}z)^T G(G^{-1}z) + (G^{-1}z)^T G(G^{-1}z)\right]$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1\alpha_2})$:

$$-\frac{1}{2}\sum_{j,\alpha_1\alpha_2} \Lambda_j^{\alpha_1}\Lambda_j^{\alpha_2} G_{\alpha_1\alpha_2} + i\sum_{j,\alpha}\Lambda_j^\alpha z_{j;\alpha}$$

$$= -\frac{1}{2}\left[\Lambda^T G\Lambda - 2i\Lambda^T z\right]$$

$$= -\frac{1}{2}\left[\Lambda^T G\Lambda - i\Lambda^T G G^{-1} z - iz^T G^{-T} G\Lambda\right]$$

$$= -\frac{1}{2}\left[\Lambda^T G\Lambda - i\Lambda^T G G^{-1} z - i(G^{-1}z)^T G\Lambda - (G^{-1}z)^T G(G^{-1}z) + (G^{-1}z)^T G(G^{-1}z)\right]$$

$$= -\frac{1}{2}\left[\Lambda^T G(\Lambda - iG^{-1}z) - i(G^{-1}z)^T G(\Lambda - G^{-1}z) + z^T G^{-1} z\right]$$

Completando quadrados mais uma vez na exponencial, reescrevendo em notação vetorial com $\Lambda = (\Lambda_j^\alpha)$; $z = (z_{j;\alpha})$; $G = (\delta_{i_1 i_2} G_{\alpha_1 \alpha_2})$:

$$
\begin{gathered}
-\frac{1}{2} \sum_{j,\alpha_1\alpha_2} \Lambda_j^{\alpha_1} \Lambda_j^{\alpha_2} G_{\alpha_1\alpha_2} + i \sum_{j,\alpha} \Lambda_j^\alpha z_{j;\alpha} \\
= -\frac{1}{2} \left[\Lambda^T G \Lambda - 2i \Lambda^T z \right] \\
= -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i z^T G^{-T} G \Lambda \right] \\
= -\frac{1}{2} \left[\Lambda^T G \Lambda - i \Lambda^T G G^{-1} z - i (G^{-1} z)^T G \Lambda - (G^{-1} z)^T G (G^{-1} z) + (G^{-1} z)^T G (G^{-1} z) \right] \\
= -\frac{1}{2} \left[\Lambda^T G (\Lambda - i G^{-1} z) - i (G^{-1} z)^T G (\Lambda - G^{-1} z) + z^T G^{-1} z \right] \\
= -\frac{1}{2} \left[(\Lambda - i G^{-1} z)^T G (\Lambda - i G^{-1} z) + z^T G^{-1} z \right].
\end{gathered}
$$

Ou seja,

$$-\frac{1}{2}\Lambda^T G\Lambda + i\Lambda^T z = -\frac{1}{2}\left[(\Lambda - iG^{-1}z)^T G(\Lambda - iG^{-1}z) + z^T G^{-1} z\right].$$

Em notação de somatório, para ficar igual ao livro

$$-\frac{1}{2}\left[(\Lambda - iG^{-1}z)^T G(\Lambda - iG^{-1}z) + z^T G^{-1} z\right]$$
$$= -\frac{1}{2}\sum_{j,\alpha_1,\alpha_2}\left[G^{(1)}_{\alpha_1\alpha_2}\left(\Lambda_{\alpha_1 j} - i\sum_{\beta_1} G_{(1)}^{\alpha_1\beta_1} z^{(1)}_{j;\beta_1}\right)\left(\Lambda_{\alpha_2 j} - i\sum_{\beta_2} G_{(1)}^{\alpha_2\beta_2} z^{(1)}_{j;\beta_2}\right) + G_{(1)}^{\alpha_1\alpha_2} z^{(1)}_{j;\alpha_1} z^{(1)}_{j;\alpha_2}\right]$$

Assim,

$$
\begin{aligned}
p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi}\right] \exp\left[(\Lambda - iG^{-1}z)^T G(\Lambda - iG^{-1}z)\right] \exp\left[z^T G^{-1} z\right] \\
&= |2\pi G^{-1}|^{\frac{n_1}{2}} \exp\left[z^T G^{-1} z\right] \\
&= \frac{1}{|2\pi G^{(1)}|^{\frac{n_1}{2}}} \exp\left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)}\right]
\end{aligned}
$$

voltando ao final o sobrescrito e a notação de somatório.

Assim,

$$
\begin{aligned}
p(z \mid D) &= \int \left[\prod_{j,\alpha} \frac{d\Lambda_{j\alpha}}{2\pi}\right] \exp\left[(\Lambda - iG^{-1}z)^T G(\Lambda - iG^{-1}z)\right] \exp\left[z^T G^{-1} z\right] \\
&= \left|2\pi G^{-1}\right|^{\frac{n_1}{2}} \exp\left[z^T G^{-1} z\right] \\
&= \frac{1}{\left|2\pi G^{(1)}\right|^{\frac{n_1}{2}}} \exp\left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2} G_{(1)}^{\alpha_1\alpha_2} z_{j;\alpha_1}^{(1)} z_{j;\alpha_2}^{(1)}\right]
\end{aligned}
$$

voltando ao final o sobrescrito e a notação de somatório.

Ou seja, o mesmo termo que encontramos na seção “Wick this way”.

Ideia: computar dois valores importantes para a seção seguinte:

Ideia: computar dois valores importantes para a seção seguinte:

- $\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right]$;

Ideia: computar dois valores importantes para a seção seguinte:

- $\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right]$;
- $\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\sigma\left(z^{(1)}_{i_2;\alpha_3}\right)\sigma\left(z^{(1)}_{i_2;\alpha_4}\right)\right]$.

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right]$$
$$=\int\left[\prod_{i=1}^{n_1}\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]$$

$$
\begin{aligned}
&\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right] \\
&= \int\left[\prod_{i=1}^{n_1}\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\times\exp\left(-\frac{1}{2}\sum_{j=1}^{n_1}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{j;\beta_1}z_{j;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2})
\end{aligned}
$$

$$
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&= \left\{\prod_{i\neq i_1}\int\left[\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left[-\frac{1}{2}\sum_{\beta_1,\beta_2\in D} G^{(1)}_{\beta_1\beta_2} z_{i;\beta_1} z_{i;\beta_2}\right]\right\}
\end{aligned}
$$

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right]$$

$$= \int\left[\prod_{i=1}^{n_1}\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\times\exp\left(-\frac{1}{2}\sum_{j=1}^{n_1}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{j;\beta_1}z_{j;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2})$$

$$= \left\{\prod_{i\neq i_1}\int\left[\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left[-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{i;\beta_1}z_{i;\beta_2}\right]\right\}$$

$$\times\int\left[\prod_{\alpha\in D}\frac{dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1;\beta_1}z_{i_1;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2})$$

$$
\begin{aligned}
&\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\right] \\
&= \int\left[\prod_{i=1}^{n_1}\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\times\exp\left(-\frac{1}{2}\sum_{j=1}^{n_1}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{j;\beta_1}z_{j;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2}) \\
&= \left\{\prod_{i\neq i_1}\int\left[\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left[-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{i;\beta_1}z_{i;\beta_2}\right]\right\} \\
&\times\int\left[\prod_{\alpha\in D}\frac{dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1;\beta_1}z_{i_1;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2}) \\
&= \{1\}\times\left[\int\prod_{\alpha\in D}\frac{dz_{i_1\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1\beta_1}z_{i_1\beta_2}\right)\sigma(z_{i_1\alpha_1})\sigma(z_{i_1\alpha_2})
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&= \left\{\prod_{i\neq i_1}\int\left[\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left[-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{(1)}_{\beta_1\beta_2}z_{i;\beta_1}z_{i;\beta_2}\right]\right\} \\
&\times\int\left[\prod_{\alpha\in D}\frac{dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1;\beta_1}z_{i_1;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2}) \\
&= \{1\}\times\left[\int\prod_{\alpha\in D}\frac{dz_{i_1\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1\beta_1}z_{i_1\beta_2}\right)\sigma(z_{i_1\alpha_1})\sigma(z_{i_1\alpha_2}) \\
&:= \langle\sigma(z_{i_1\alpha_1})\sigma(z_{i_1\alpha_2})\rangle_{G^{(1)}}.
\end{aligned}
$$

Com a expressão do final, reintroduzimos a notação para o correlator de $F(z_{\alpha_1}, \ldots, z_{\alpha_m})$ com matriz de variância $g = (g_{\alpha_1 \alpha_2})_{N_D \times N_D}$:

$$\langle F(z_{\alpha_1}, \ldots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_\alpha}{\sqrt{|2\pi g|}} \right] \exp\left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \ldots, z_{\alpha_m})$$

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e denotamos ainda:

$$\sigma_\alpha := \sigma(z_\alpha)$$

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e denotamos ainda:

$$\sigma_\alpha := \sigma(z_\alpha)$$

Assim, podemos escrever

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1} \right) \sigma\left(z^{(1)}_{i_1;\alpha_2} \right) \right] = \left\langle \sigma_{\alpha_1} \sigma_{\alpha_2} \right\rangle_{G^{(1)}}$$

É possível ainda generalizar para correlatores de mais pontos. Para todas as saídas do mesmo neurônio i_1:

$$\mathbb{E}\left[\sigma(z^{(1)}_{i_1;\alpha_1})\sigma(z^{(1)}_{i_1;\alpha_2})\sigma(z^{(1)}_{i_1;\alpha_3})\sigma(z^{(1)}_{i_1;\alpha_4})\right] = \langle\sigma_{\alpha_1}\sigma_{\alpha_2}\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}}$$

fazendo as exatas mesmas etapas.

Para $i_1 \neq i_2$

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\sigma\left(z^{(1)}_{i_2;\alpha_3}\right)\sigma\left(z^{(1)}_{i_2;\alpha_4}\right)\right]$$

Para $i_1 \neq i_2$

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\sigma\left(z^{(1)}_{i_2;\alpha_3}\right)\sigma\left(z^{(1)}_{i_2;\alpha_4}\right)\right]$$

$$= \left\{\prod_{i\notin\{i_1,i_2\}} \int \left[\prod_{\alpha\in D} \frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right] \exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D} G^{\beta_1\beta_2}_{(1)} z_{i;\beta_1} z_{i;\beta_2}\right)\right\}$$

$$\times \int \left[\frac{\prod_{\alpha\in D} dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right] \exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D} G^{\beta_1\beta_2}_{(1)} z_{i_1;\beta_1} z_{i_1;\beta_2}\right) \sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2})$$

$$\times \int \left[\frac{\prod_{\alpha\in D} dz_{i_2;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right] \exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D} G^{\beta_1\beta_2}_{(1)} z_{i_2;\beta_1} z_{i_2;\beta_2}\right) \sigma(z_{i_2;\alpha_3})\sigma(z_{i_2;\alpha_4})$$

Para $i_1 \neq i_2$

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\sigma\left(z^{(1)}_{i_2;\alpha_3}\right)\sigma\left(z^{(1)}_{i_2;\alpha_4}\right)\right]$$

$$= \left\{\prod_{i\notin\{i_1,i_2\}}\int\left[\prod_{\alpha\in D}\frac{dz_{i;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i;\beta_1}z_{i;\beta_2}\right)\right\}$$

$$\times\int\left[\frac{\prod_{\alpha\in D}dz_{i_1;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_1;\beta_1}z_{i_1;\beta_2}\right)\sigma(z_{i_1;\alpha_1})\sigma(z_{i_1;\alpha_2})$$

$$\times\int\left[\frac{\prod_{\alpha\in D}dz_{i_2;\alpha}}{\sqrt{|2\pi G^{(1)}|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D}G^{\beta_1\beta_2}_{(1)}z_{i_2;\beta_1}z_{i_2;\beta_2}\right)\sigma(z_{i_2;\alpha_3})\sigma(z_{i_2;\alpha_4})$$

$$= \left\langle\sigma(z_{\alpha_1})\sigma(z_{\alpha_2})\right\rangle_{G^{(1)}}\left\langle\sigma(z_{\alpha_3})\sigma(z_{\alpha_4})\right\rangle_{G^{(1)}}$$

MTM

I.e.:

$$\mathbb{E}\left[\sigma\left(z^{(1)}_{i_1;\alpha_1}\right)\sigma\left(z^{(1)}_{i_1;\alpha_2}\right)\sigma\left(z^{(1)}_{i_2;\alpha_3}\right)\sigma\left(z^{(1)}_{i_2;\alpha_4}\right)\right] = \left\langle\sigma(z_{\alpha_1})\sigma(z_{\alpha_2})\right\rangle_{G^{(1)}}\left\langle\sigma(z_{\alpha_3})\sigma(z_{\alpha_4})\right\rangle_{G^{(1)}}$$

Isso ilustra a independência entre as distribuições de neurônios diferentes.


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NORMAL

Lembrando, denotamos:

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As entradas da segunda camada são dadas por:

$$z_{i,\alpha}^{(2)} \equiv z_i^{(2)}(x_\alpha) = b_i^{(2)} + \sum_{j=1}^{n_1} W_{ij}^{(2)}\, \sigma_{j;\alpha}^{(1)}, \quad \text{para } i = 1, \ldots, n_2. \tag{2}$$

A distribuição conjunta da segunda e primeira camada pode ser dada por::

$$p(z^{(2)}, z^{(1)} \mid D) = p(z^{(2)} \mid z^{(1)}) p(z^{(1)} \mid D) \tag{3}$$

1º: $p(z^{(1)} \mid D)$ - Seção 4.1;

A distribuição conjunta da segunda e primeira camada pode ser dada por::

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1º: $p(z^{(1)} \mid D)$ - Seção 4.1;

2º: Como definido na seção 2:

$$p(z^{(2)} \mid z^{(1)}) = \int \left[\prod_i db_i^{(2)} p(b_i^{(2)}) \right] \left[\prod_{i,j} dW_{ij}^{(2)} p(W_{ij}^{(2)}) \right] \prod_{i,\alpha} \delta \left(z_i^{(2)}(x_\alpha) - b_i^{(2)} - \sum_{j=1}^{n_1} W_{ij}^{(2)} \sigma_{j;\alpha}^{(1)} \right) \tag{4}$$

Ainda, podemos calcular as probabilidades marginalizadas como:

$$p(z^{(2)} \mid D) = \int \left[\prod_{i;\alpha} dz_{i;\alpha}^{(1)} \right] p(z^{(1)} \mid z^{(2)}) p(z^{(1)} \mid D). \qquad (5)$$

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Desse modo, a probabilidade fica:

$$p(z^{(2)} \mid z^{(1)}) = \frac{1}{|2\pi \widehat{G}^{(2)}|^{\frac{n_2}{2}}} \exp\left[-\frac{1}{2} \sum_{j,\alpha_1,\alpha_2}^{n_2} \widehat{G}^{\alpha_1\alpha_2}_{(2)} z^{(2)}_{j;\alpha_1} z^{(2)}_{j;\alpha_2}\right] \quad (6)$$

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onde definimos a **métrica *estocástica* de segunda camada**:

$$\widehat{G}^{(2)}_{\alpha_1\alpha_2} := C^{(2)}_b + C^{(2)}_W \frac{1}{n_1} \sum_{j=1}^{n_1} \sigma_{j;\alpha_1} \sigma_{j;\alpha_2} \quad (7)$$

Importante ressaltar que a variável $\widehat{G}^{(2)}$ é *estocástica*:

$$\widehat{G}^{(2)}_{\alpha_1\alpha_2} := C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \sigma_{j;\alpha_1} \sigma_{j;\alpha_2} \tag{8}$$

pois depende da variável aleatória $\sigma^{(1)} = \sigma(z^{(1)})$.

Podemos então calcular seu valor esperado:

$$\mathbb{E}[\widehat{G}^{(2)}_{\alpha_1\alpha_2}] = C_b^{(2)} + C_W^{(2)} \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbb{E}[\sigma_{j;\alpha_1} \sigma_{j;\alpha_2}]$$

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Chamamos o valor $G^{(2)}_{\alpha_1\alpha_2}$ de **métrica média de 2ª camada.**

Definimos ainda **flutuação de 2ª camada**:

$$\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2} := \widehat{G}^{(2)}_{\alpha_1\alpha_2} - G^{(2)}_{\alpha_1\alpha_2}$$

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Note que por construção, $\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}$ tem média 0 sobre a primeira camada de ativações:

$$\mathbb{E}[\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}] = \mathbb{E}[\widehat{G}^{(2)}_{\alpha_1\alpha_2}] - \mathbb{E}[\mathbb{E}[\widehat{G}^{(2)}_{\alpha_1\alpha_2}]] = 0$$

Estamos interessados no correlator de 2 pontos de $\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}$:

$$\mathbb{E}\left[\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}\widehat{\Delta G}^{(2)}_{\alpha_3\alpha_4}\right]$$

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$$= \left(\frac{C_W^{(2)}}{n_1}\right)^2 \sum_{j,k=1}^{n_1} \mathbb{E}\left[\left(\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2} - \mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}]\right)\left(\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4} - \mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}]\right)\right]$$

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Temos:

$$\mathbb{E}\left[\left(\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}-\mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}]\right)\left(\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}-\mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}]\right)\right]$$

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&= \mathbb{E}\left[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}\right] - \mathbb{E}\left[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}\right]\mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}] \\
&\quad - \mathbb{E}[\sigma^{(1)}_{j;\alpha_3}\sigma^{(1)}_{j;\alpha_4}]\mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}] + \mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}]\mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}]
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&\mathbb{E}\left[\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}\widehat{\Delta G}^{(2)}_{\alpha_3\alpha_4}\right] \\
&= \left(\frac{C_W^{(2)}}{n_1}\right)^2 \sum_{j,k=1}^{n_1} \mathbb{E}\left[\left(\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2} - \mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}]\right)\left(\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4} - \mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}]\right)\right]
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\end{aligned}
$$

Note que, para $j \neq k$,

$$\mathbb{E}\left[\sigma_{j;\alpha_1}^{(1)}\sigma_{j;\alpha_2}^{(1)}\sigma_{k;\alpha_3}^{(1)}\sigma_{k;\alpha_4}^{(1)}\right] = \mathbb{E}\left[\sigma_{j;\alpha_1}^{(1)}\sigma_{j;\alpha_2}^{(1)}\right]\left[\sigma_{j;\alpha_3}^{(1)}\sigma_{j;\alpha_4}^{(1)}\right]$$

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Com isso,

$$\left(\frac{C_W^{(2)}}{n_1}\right)^2 \sum_{j,k=1}^{n_1} \mathbb{E}\left[\left(\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2} - \mathbb{E}[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}]\right)\left(\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4} - \mathbb{E}[\sigma^{(1)}_{k;\alpha_3}\sigma^{(1)}_{k;\alpha_4}]\right)\right]$$

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&= \left(\frac{C_W^{(2)}}{n_1}\right)^2 \sum_{j=1}^{n_1} \left\{\mathbb{E}\left[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}\sigma^{(1)}_{j;\alpha_3}\sigma^{(1)}_{j;\alpha_4}\right] - \mathbb{E}\left[\sigma^{(1)}_{j;\alpha_1}\sigma^{(1)}_{j;\alpha_2}\right]\mathbb{E}\left[\sigma^{(1)}_{j;\alpha_3}\sigma^{(1)}_{j;\alpha_4}\right]\right\} \\
&= \left(\frac{C_W^{(2)}}{n_1}\right)^2 \sum_{j=1}^{n_1} \left\{\langle\sigma_{\alpha_1}\sigma_{\alpha_2}\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}} - \langle\sigma_{\alpha_1}\sigma_{\alpha_2}\rangle_{G^{(1)}}\langle\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}}\right\} \\
&= \frac{1}{n_1}\left(C_W^{(2)}\right)^2 \left[\langle\sigma_{\alpha_1}\sigma_{\alpha_2}\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}} - \langle\sigma_{\alpha_1}\sigma_{\alpha_2}\rangle_{G^{(1)}}\langle\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}}\right]
\end{aligned}$$

O valor

$$V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} := \left(C_W^{(2)}\right)^2 \left[\langle\sigma_{\alpha_1}\sigma_{\alpha_2}\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}} - \langle\sigma_{\alpha_1}\sigma_{\alpha_2}\rangle_{G^{(1)}}\langle\sigma_{\alpha_3}\sigma_{\alpha_4}\rangle_{G^{(1)}}\right]$$

é chamado de **vértice de quatro pontos** de 2ª camada.

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é chamado de **vértice de quatro pontos** de 2ª camada. Desse modo, podemos escrever

$$\mathbb{E}\left[\widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}\widehat{\Delta G}^{(2)}_{\alpha_3\alpha_4}\right] = \frac{1}{n_1} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}.$$

Observações:

- $V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} == V(x_{\alpha_1}, x_{\alpha_2}; x_{\alpha_3}, x_{\alpha_4});$
- O vértice de quatro pontos é simétrico sobre $\alpha_1 \leftrightarrow \alpha_2$, $\alpha_3 \leftrightarrow \alpha_4$, $(\alpha_1, \alpha_2) \leftrightarrow (\alpha_3, \alpha_4)$.

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Observações:

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    - $V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} = V^{(2)}_{(\alpha_3\alpha_4)(\alpha_1\alpha_2)}$;
- Quando $n_1 \gg 1$, $\frac{1}{n_1} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} \to 0$, de modo que a distribuição se aproxima de uma Gaussiana.

Como visto na Equação (6), $p(z^{(2)} \mid z^{(1)})$ é gaussiana (em relação a $z^{(1)}$), com média 0 e variância (estocástica) $\widehat{G}^{(2)}_{\alpha_1\alpha_2}$:

$$p(z^{(2)} \mid z^{(1)}) = \frac{1}{|2\pi\widehat{G}^{(2)}|^{\frac{n_2}{2}}} \exp\left[-\frac{1}{2}\sum_{j,\alpha_1,\alpha_2}^{n_2} \widehat{G}^{\alpha_1\alpha_2}_{(2)} z^{(2)}_{j;\alpha_1} z^{(2)}_{j;\alpha_2}\right]$$

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Isso nos permite usar as *contrações de Wick*.

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Isso nos permite usar as *contrações de Wick*. Para o correlator de 2 pontos:

$$\begin{aligned}
\mathbb{E}\left[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2}\right] &= \delta_{i_1 i_2}\mathbb{E}[\widehat{G}^{(2)}_{\alpha_1\alpha_2}] \\
&= \delta_{i_1 i_2} G^{(2)}_{\alpha_1\alpha_2} \\
&= \delta_{i_1 i_2}\left(C^{(2)}_b + C^{(2)}_W \langle \sigma_{\alpha_1}\sigma_{\alpha_2}\rangle_{G^{(1)}}\right)
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Obs: o livro não explica, mas o valor esperado deve aparecer aqui no teorema de Wick pela variância ser uma variável estocástica. Uma conta parecida com a Seção 4.1 (Wick this way) também revela onde surge a esperança.

No correlator de 4 pontos:

$$\mathbb{E}\big[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}\big]$$
$$= \delta_{i_1 i_2}\delta_{i_3 i_4}\,\mathbb{E}\big[\widehat{G}^{(2)}_{\alpha_1\alpha_2}\widehat{G}^{(2)}_{\alpha_3\alpha_4}\big] + \delta_{i_1 i_3}\delta_{i_2 i_4}\,\mathbb{E}\big[\widehat{G}^{(2)}_{\alpha_1\alpha_3}\widehat{G}^{(2)}_{\alpha_2\alpha_4}\big] + \delta_{i_1 i_4}\delta_{i_2 i_3}\,\mathbb{E}\big[\widehat{G}^{(2)}_{\alpha_1\alpha_4}\widehat{G}^{(2)}_{\alpha_2\alpha_3}\big]$$

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Lembrando,

- $\widehat{G}^{(2)}_{\alpha_1\alpha_2} = G^{(2)}_{\alpha_1\alpha_2} + \widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2};$

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Assim:

$$\mathbb{E}\left[\widehat{G}^{(2)}_{\alpha_1\alpha_2}\widehat{G}^{(2)}_{\alpha_3\alpha_4}\right] = \mathbb{E}\left[\left(G^{(2)}_{\alpha_1\alpha_2} + \widehat{\Delta G}^{(2)}_{\alpha_1\alpha_2}\right)\left(G^{(2)}_{\alpha_3\alpha_4} + \widehat{\Delta G}^{(2)}_{\alpha_3\alpha_4}\right)\right]$$

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$$= G^{(2)}_{\alpha_1\alpha_2}G^{(2)}_{\alpha_3\alpha_4} + V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}$$

O mesmo argumento se repete para as outras permutações de índices.

Com isso,

$$
\begin{aligned}
&\mathbb{E}\big[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}\big] \\
&= \delta_{i_1 i_2}\delta_{i_3 i_4}\,\mathbb{E}\big[\hat{G}^{(2)}_{\alpha_1\alpha_2}\hat{G}^{(2)}_{\alpha_3\alpha_4}\big] + \delta_{i_1 i_3}\delta_{i_2 i_4}\,\mathbb{E}\big[\hat{G}^{(2)}_{\alpha_1\alpha_3}\hat{G}^{(2)}_{\alpha_2\alpha_4}\big] + \delta_{i_1 i_4}\delta_{i_2 i_3}\,\mathbb{E}\big[\hat{G}^{(2)}_{\alpha_1\alpha_4}\hat{G}^{(2)}_{\alpha_2\alpha_3}\big]
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&= \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(2)}_{\alpha_1\alpha_2} G^{(2)}_{\alpha_3\alpha_4} + \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(2)}_{\alpha_1\alpha_3} G^{(2)}_{\alpha_2\alpha_4} + \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(2)}_{\alpha_1\alpha_4} G^{(2)}_{\alpha_2\alpha_3} \\
&+ \frac{1}{n_1}\Big[\delta_{i_1 i_2}\delta_{i_3 i_4} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} + \delta_{i_1 i_3}\delta_{i_2 i_4} V^{(2)}_{(\alpha_1\alpha_3)(\alpha_2\alpha_4)} + \delta_{i_1 i_4}\delta_{i_2 i_3} V^{(2)}_{(\alpha_1\alpha_4)(\alpha_2\alpha_3)}\Big]
\end{aligned}
$$

Isso nos permite ver que:

$$
\begin{aligned}
&\mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}]_{\text{connected}} \\
&= \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] \\
&- \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2}]\mathbb{E}[z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_3;\alpha_3}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_4;\alpha_4}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3}]
\end{aligned}
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&= \mathbb{E}\big[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}\big]\\
&- \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(2)}_{\alpha_1\alpha_2} G^{(2)}_{\alpha_3\alpha_4} - \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(2)}_{\alpha_1\alpha_3} G^{(2)}_{\alpha_2\alpha_4} - \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(2)}_{\alpha_1\alpha_4} G^{(2)}_{\alpha_2\alpha_3}
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&- \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2}]\mathbb{E}[z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_3;\alpha_3}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_4;\alpha_4}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3}] \\
&= \mathbb{E}\big[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}\big] \\
&- \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(2)}_{\alpha_1\alpha_2} G^{(2)}_{\alpha_3\alpha_4} - \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(2)}_{\alpha_1\alpha_3} G^{(2)}_{\alpha_2\alpha_4} - \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(2)}_{\alpha_1\alpha_4} G^{(2)}_{\alpha_2\alpha_3} \\
&= \frac{1}{n_1}\Big[\delta_{i_1 i_2}\delta_{i_3 i_4} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} + \delta_{i_1 i_3}\delta_{i_2 i_4} V^{(2)}_{(\alpha_1\alpha_3)(\alpha_2\alpha_4)} + \delta_{i_1 i_4}\delta_{i_2 i_3} V^{(2)}_{(\alpha_1\alpha_4)(\alpha_2\alpha_3)}\Big]
\end{aligned}
$$

Isso nos permite ver que:

$$
\begin{aligned}
&\mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}]_{\text{connected}} \\
&= \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] \\
&- \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2}]\mathbb{E}[z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_3;\alpha_3}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_4;\alpha_4}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3}] \\
&= \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] \\
&- \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(2)}_{\alpha_1\alpha_2} G^{(2)}_{\alpha_3\alpha_4} - \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(2)}_{\alpha_1\alpha_3} G^{(2)}_{\alpha_2\alpha_4} - \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(2)}_{\alpha_1\alpha_4} G^{(2)}_{\alpha_2\alpha_3} \\
&= \frac{1}{n_1}\left[\delta_{i_1 i_2}\delta_{i_3 i_4} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} + \delta_{i_1 i_3}\delta_{i_2 i_4} V^{(2)}_{(\alpha_1\alpha_3)(\alpha_2\alpha_4)} + \delta_{i_1 i_4}\delta_{i_2 i_3} V^{(2)}_{(\alpha_1\alpha_4)(\alpha_2\alpha_3)}\right]
\end{aligned}
$$

Observações:

- Isso evidencia a importância de $V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}$ na "não-gaussianidade" de $p(z^{(2)} \mid D)$;

Isso nos permite ver que:

$$
\begin{aligned}
&\mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}]_{\text{connected}} \\
&= \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] \\
&- \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2}]\mathbb{E}[z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_3;\alpha_3}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_4;\alpha_4}] - \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_4;\alpha_4}]\mathbb{E}[z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3}] \\
&= \mathbb{E}[z^{(2)}_{i_1;\alpha_1} z^{(2)}_{i_2;\alpha_2} z^{(2)}_{i_3;\alpha_3} z^{(2)}_{i_4;\alpha_4}] \\
&- \delta_{i_1 i_2}\delta_{i_3 i_4} G^{(2)}_{\alpha_1\alpha_2} G^{(2)}_{\alpha_3\alpha_4} - \delta_{i_1 i_3}\delta_{i_2 i_4} G^{(2)}_{\alpha_1\alpha_3} G^{(2)}_{\alpha_2\alpha_4} - \delta_{i_1 i_4}\delta_{i_2 i_3} G^{(2)}_{\alpha_1\alpha_4} G^{(2)}_{\alpha_2\alpha_3} \\
&= \frac{1}{n_1}\Big[\delta_{i_1 i_2}\delta_{i_3 i_4} V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} + \delta_{i_1 i_3}\delta_{i_2 i_4} V^{(2)}_{(\alpha_1\alpha_3)(\alpha_2\alpha_4)} + \delta_{i_1 i_4}\delta_{i_2 i_3} V^{(2)}_{(\alpha_1\alpha_4)(\alpha_2\alpha_3)}\Big]
\end{aligned}
$$

Observações:

- Isso evidencia a importância de $V^{(2)}_{(\alpha_1\alpha_2)(\alpha_3\alpha_4)}$ na "não-gaussianidade" de $p(z^{(2)} \mid D)$;
- Mais uma vez mostra que se $n_1 \to \infty$, o correlator se aproxima de 0 e $p(z^{(2)} \mid D)$ se aproxima de uma gaussiana.

Vamos atrás de uma ação quártica:

$$S[z] = \frac{1}{2} \sum_{\alpha_1 \alpha_2 \in D} g^{\alpha_1 \alpha_2} \sum_{i=1}^{n} z_{i;\alpha_1} z_{i;\alpha_2} \\ - \frac{1}{8} \sum_{\alpha_1, \ldots, \alpha_4 \in D} v^{(\alpha_1 \alpha_2)(\alpha_3 \alpha_4)} \sum_{i_1, i_2 = 1}^{n} z_{i_1;\alpha_1} z_{i_1;\alpha_2} z_{i_2;\alpha_3} z_{i_2;\alpha_4}$$

com g e v por hora indeterminados.

Em 4.1, lembramos a notação

$$\langle F(z_{\alpha_1}, \ldots, z_{\alpha_m}) \rangle_g \equiv \int \left[\frac{\prod_{\alpha \in D} dz_\alpha}{\sqrt{|2\pi g|}} \right] \exp\left(-\frac{1}{2} \sum_{\beta_1, \beta_2 \in D} g^{\beta_1 \beta_2} z_{\beta_1} z_{\beta_2} \right) F(z_{\alpha_1}, \ldots, z_{\alpha_m})$$

Em 4.1, lembramos a notação

$$\langle F(z_{\alpha_1},\ldots,z_{\alpha_m})\rangle_g \equiv \int \left[\frac{\prod_{\alpha\in D} dz_\alpha}{\sqrt{|2\pi g|}}\right]\exp\left(-\frac{1}{2}\sum_{\beta_1,\beta_2\in D} g^{\beta_1\beta_2} z_{\beta_1} z_{\beta_2}\right) F(z_{\alpha_1},\ldots,z_{\alpha_m})$$

Agora, introduzimos a notação:

$$\begin{aligned}&\langle\!\langle F(z_{i_1;\alpha_1},\ldots,z_{i_m;\alpha_m})\rangle\!\rangle_g\\ &= \left[\prod_{i=1}^{n}\frac{\prod_{\alpha\in D} dz_{i;\alpha}}{\sqrt{|2\pi g|}}\right]\exp\left(-\frac{1}{2}\sum_{j=1}^{n}\sum_{\beta_1,\beta_2\in D} g^{(\beta_1\beta_2)} z_{j;\beta_1} z_{j;\beta_2}\right) F(z_{i_1;\alpha_1},\ldots,z_{i_m;\alpha_m})\end{aligned}$$

Obtemos então:

$$\mathbb{E}[F(z_{i_1\alpha_1},\dots,z_{i_m\alpha_m})]$$

$$= \frac{\int \prod_{i;\alpha} dz_{i;\alpha}\, e^{-S(z)} F(z_{i_1;\alpha_1},\dots,z_{i_m;\alpha_m})}{\int \prod_{i;\alpha} dz_{i;\alpha}\, e^{-S(z)}}$$

Obtemos então:

$$\mathbb{E}[F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})]$$
$$= \frac{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1},\ldots,z_{i_m;\alpha_m})}{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)}}$$
$$= \frac{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\} F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g}{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\}\rangle\!\rangle_g}$$

Obtemos então:

$$\mathbb{E}[F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})]$$
$$= \frac{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1},\ldots,z_{i_m;\alpha_m})}{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)}}$$
$$= \frac{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\} F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g}{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\}\rangle\!\rangle_g}$$
$$\langle\!\langle F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g$$
$$+ \frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} [\langle\!\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g$$
$$- \langle\!\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\rangle\!\rangle_g \langle\!\langle F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g]$$
$$+ O(v^2)$$

Obtemos então:

$$\mathbb{E}[F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})]$$

$$= \frac{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)} F(z_{i_1;\alpha_1},\ldots,z_{i_m;\alpha_m})}{\int \prod_{i;\alpha} dz_{i;\alpha} e^{-S(z)}}$$

$$= \frac{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\} F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g}{\langle\!\langle \exp\left\{\frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\right\}\rangle\!\rangle_g}$$

$$\langle\!\langle F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g$$

$$+ \frac{1}{8}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)} \sum_{j_1,j_2=1}^{n} [\langle\!\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4} F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g$$

$$- \langle\!\langle z_{j_1;\beta_1} z_{j_1;\beta_2} z_{j_2;\beta_3} z_{j_2;\beta_4}\rangle\!\rangle_g \langle\!\langle F(z_{i_1\alpha_1},\ldots,z_{i_m\alpha_m})\rangle\!\rangle_g]$$

$$+ O(v^2)$$

Obs: no final, usando a expansão de Taylor $e^v \approx 1 + v + O(v^2)$.

Por exemplo, no correlator de dois pontos,

$$
\begin{aligned}
&\mathbb{E}\left[z_{i_1;\alpha_1}\, z_{i_2;\alpha_2}\right] \\
&= \delta_{i_1 i_2}\left[g_{\alpha_1\alpha_2} + \frac{1}{2}\sum_{\beta_1,\ldots,\beta_4\in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)}\left(n\, g_{\alpha_1\beta_1}\, g_{\alpha_2\beta_2}\, g_{\beta_3\beta_4} + 2\, g_{\alpha_1\beta_1}\, g_{\alpha_2\beta_3}\, g_{\beta_2\beta_4}\right)\right] + \mathcal{O}(v^2)
\end{aligned}
$$

sendo $g_{\alpha_1\alpha_2}$ a inversa do termo quadrático $g^{\alpha_1\alpha_2}$ de $S[z]$.

Já no correlator de 4 pontos,

$$
\begin{aligned}
&\mathbb{E}\left[z_{i_1;\alpha_1} z_{i_2;\alpha_2} z_{i_3;\alpha_3} z_{i_4;\alpha_4}\right]_{\text{connected}} \\
&\equiv \mathbb{E}\left[z_{i_1;\alpha_1} z_{i_2;\alpha_2} z_{i_3;\alpha_3} z_{i_4;\alpha_4}\right] - \mathbb{E}\left[z_{i_1;\alpha_1} z_{i_2;\alpha_2}\right]\mathbb{E}\left[z_{i_3;\alpha_3} z_{i_4;\alpha_4}\right] \\
&\quad - \mathbb{E}\left[z_{i_1;\alpha_1} z_{i_3;\alpha_3}\right]\mathbb{E}\left[z_{i_2;\alpha_2} z_{i_4;\alpha_4}\right] - \mathbb{E}\left[z_{i_1;\alpha_1} z_{i_4;\alpha_4}\right]\mathbb{E}\left[z_{i_2;\alpha_2} z_{i_3;\alpha_3}\right] \\
&= \delta_{i_1 i_2}\delta_{i_3 i_4} \sum_{\beta_1,\ldots,\beta_4 \in D} v^{(\beta_1\beta_2)(\beta_3\beta_4)}\, g_{\alpha_1\beta_1}\, g_{\alpha_2\beta_2}\, g_{\alpha_3\beta_3}\, g_{\alpha_4\beta_4} \\
&\quad + \delta_{i_1 i_3}\delta_{i_2 i_4} \sum_{\beta_1,\ldots,\beta_4 \in D} v^{(\beta_1\beta_3)(\beta_2\beta_4)}\, g_{\alpha_1\beta_1}\, g_{\alpha_3\beta_3}\, g_{\alpha_2\beta_2}\, g_{\alpha_4\beta_4} \\
&\quad + \delta_{i_1 i_4}\delta_{i_2 i_3} \sum_{\beta_1,\ldots,\beta_4 \in D} v^{(\beta_1\beta_4)(\beta_2\beta_3)}\, g_{\alpha_1\beta_1}\, g_{\alpha_4\beta_4}\, g_{\alpha_2\beta_2}\, g_{\alpha_3\beta_3} \\
&\quad + \mathcal{O}(v^2).
\end{aligned}
$$

Comparando os slides anteriores com as contas feitas na subseção anterior, estabelecendo:

$$g^{\alpha_1\alpha_2} = G_{(2)}^{\alpha_1\alpha_2} + O(\frac{1}{n_1})$$
$$v^{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} = \frac{1}{n_1}V_{(2)}^{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} + O(\frac{1}{n_1^2})$$

reproduzimos os correlatores de 2ª camada até ordem $\frac{1}{n_1}$, com a distribuição marginal

$$p(z^{(2)} \mid D) = \frac{1}{Z}e^{-S[z^{(2)}]}$$

Comparando os slides anteriores com as contas feitas na subseção anterior, estabelecendo:

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reproduzimos os correlatores de 2ª camada até ordem $\frac{1}{n_1}$, com a distribuição marginal

$$p(z^{(2)} \mid D) = \frac{1}{Z}e^{-S[z^{(2)}]}$$

Obs: aqui,

$$V_{(2)}^{(\alpha_1\alpha_2)(\alpha_3\alpha_4)} = \sum_{\beta_1,\ldots,\beta_4} G_{(2)}^{\alpha_1\beta_1}G_{(2)}^{\alpha_2\beta_2}G_{(2)}^{\alpha_3\beta_3}G_{(2)}^{\alpha_4\beta_4}V^{(2)}_{(\beta_1\beta_2)(\beta_3\beta_4)}$$

Obrigado!

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Repositório com os materiais da apresentação:
https://github.com/felipekriffel/Mestrado